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Original Paper

Full separation of oligomers in block copolymers of ethylene oxide and propylene oxide

In this study di- and triblock copolymers of ethylene oxide (EO) and propylene oxide (PO) are analyzed by using CAP for one block and adsorption for the other block. This gives a complete picture of EO- and PO-based block copolymer with respect to the oligomers of both blocks. A full resolution of the individual oligomers can be achieved for both blocks up to an average molecular weight of 1000–1500 of each block.

Keywords: Amphiphilic block copolymers / Critical adsorption point (CAP) / Liquid chromatography under critical conditions (LCCC) / Oligomers

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1 Introduction

Amphiphilic block copolymers are a very important group of modern polymeric materials. Along with applications in many different fields like nano- and biomaterials [1], modification of surface to control protein adsorption [1], for controlled release [2], for nanofiltration membranes, as synthetic lubricants and emulsifiers and for biomedical applications; these fascinating materials bring many new challenges to polymer chemists and technologists [3, 4].

An especially important class of block copolymers are block copolymers of ethylene oxide (EO) and propylene oxide (PO), which have a very wide range of applications [4].

Basically, such materials can consist of two or three blocks. In the triblocks, the blocks may be arranged in a different way: the hydrophilic block (EO) may form the center of the chain or the arms. In the following text, we shall use the abbreviations EO–PO for the diblock copolymer and EO–PO–EO or PO–EO–PO for the triblock copolymers.

The diblocks are typically synthesized by propoxylation of monoethers of PEG, while the triblocks can be obtained by ethoxylation of polypropylene glycol or by propoxylation of PEG. As possible side products, the tri-

blocks may contain the homopolymers as well as the diblock copolymer.

Obviously, the physical properties of the block copolymers are strongly influenced by the distributions of molar mass, functionality, chemical composition, architecture, and content of the side products [5]. Consequently there is a strong need for reliable analytical methods, which can provide the required information in order to improve the synthetic procedures to yield well defined products of high purity.

Basically, one may apply different modes of LC and spectroscopic techniques (such as NMR and MALDI-TOF-MS), and combinations of these. NMR alone (without any separation) yields only average values. MALDI-TOF-MS is problematic with highly polydisperse samples. When applied to fractions from a chromatographic separation, which have rather narrow distributions, it is, however, a very powerful technique.

The most popular chromatographic technique in polymer analysis is size exclusion chromatography (SEC), which yields the molar mass distribution (MMD) [6, 7]. SEC with dual detection [7–13] yields information on the chemical composition along the MMD. It must, however, be mentioned, that SEC cannot differentiate between a block copolymer and a mixture of homopolymers.

This can, however, be achieved by using LC under critical conditions (LCCC) [14–24]. At a special temperature and in a special mobile phase composition, retention is independent of the molar mass of a polymer chain or one block; which thus becomes “chromatographically invisible”. This situation is called the critical adsorption point (CAP). Under these conditions, one may separate a block copolymer from a homopolymer. At the CAP for one block, the elution volume depends on the chemical nature and the size of the other block, which may be

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Abbreviation: ACN, acetonitrile; CAP, critical adsorption point; EO, ethylene oxide; LAC, liquid adsorption chromatography; SEC, size exclusion chromatography; PO, propylene oxide

excluded from the pores or interact with the stationary phase. In the first case it elutes in the SEC regime (*i.e.*, earlier than the critical homopolymer, which elutes at the void volume of the column), in the latter case it will elute in the regime of liquid adsorption chromatography (LAC), *i.e.*, much later than the critical homopolymer [17, 24–26].

In the analysis of block copolymers, the first situation is commonly used [8–11, 24], while the second one is typically applied in the separation of functional polymers [8, 10]. Obviously, SEC for the noncritical block will work for any size of this block, provided that it is large enough to be separated from the solvent peak.

LAC of the noncritical block works only in a limited range of molar mass, as retention increases exponentially with the size of this block. At the mobile phase composition corresponding to the CAP for the first block retention may be too strong, if the adsorbed block is long. In mobile phases consisting of different solvents, the retention of one block at the CAP for the other one may be different [17]. A fine adjustment can be achieved by changing the temperature and the pore size of the column [17, 27, 28]: columns with the same chemical nature, but different pore size will have a CAP at the same mobile phase composition. As retention in LAC depends on the pore surface (and thus on the pore diameter), one may separate copolymers of different molar mass on columns with different porosity [27–30].

In the case of EO- and PO-based block copolymers, the approach with an excluded noncritical block can be applied both to EO–PO–EO and PO–EO–PO triblocks [24].

The situation with adsorbed noncritical blocks works quite well for diblocks, for triblocks it is, however, much more complicated, as will be discussed later on. In a triblock, it is only possible to make the outer block invisible and to separate the inner block by LAC, while at the CAP for the inner block retention is not independent of its length [31, 32]. This means, that in 2-D separations it makes a difference, which mechanism is used in the first and the second dimension.

In principle, LAC for PO at the CAP for EO can be achieved in RP systems [17, 25, 26].

LAC for EO at the CAP for PO should be possible on normal phase systems. Such separations have not yet been described in the literature.

In this study, we have tried to find a system, which allows an LAC separation according to the number of EO units at the CAP for PO.

2 Separation mechanisms in LC of polymers

All modes of LC can be described by a theory (using a model of a Gaussian polymer chain and a slit-like pore)

for the distribution coefficient of flexible homopolymers, which are capable of interacting with the pore walls [19, 22, 31, 33, 34]. According to this theory, the distribution coefficient K is a function of three parameters: the pore width D , the average radius of gyration R of the polymer coil and the adsorption interaction parameter c , which describes the interaction of a structural unit with stationary phase [35]. The interaction parameter is negative in SEC, where retention decreases with increasing molar mass and is positive in LAC, where retention increases exponentially with the number of repeat units.

In both cases, the elution volume V_e depends on the distribution coefficient K of the polymer between the interstitial volume V_i and the pore volume V_p

$$V_e = V_i + KV_p \quad (1)$$

This equation is well known in SEC (where $0 < K < 1$): a polymer of very high molar mass (above the exclusion limit) elutes at V_i (as $K = 0$), while small molecules (which may access the entire pore volume) elute at the void volume V_0 (as $K = 1$):

$$V_0 = V_i + V_p \quad (2)$$

In LAC, the distribution coefficient can assume values much larger than 1.

The distribution coefficient is given by the change in free energy related to the transition of the polymer from the interstitial volume into the pores of the stationary phase:

$$\ln K = \Delta G = \Delta H - T\Delta S \quad (3)$$

In general, the driving force in SEC is the entropy change, while in LAC it is the enthalpic interaction of repeat unit with the stationary phase.

In ideal SEC, where c is strongly negative, the distribution coefficient depends only on the ratio of the pore diameter D and the radius of gyration R of the polymer [36, 37].

$$K_{\text{SEC}} = \frac{V_e - V_i}{V_p} = \left(1 - \frac{4}{\sqrt{\pi}} \frac{R}{D}\right) \quad (4)$$

and the elution volume is given by

$$V_{\text{SEC}} = V_i + V_p \left(1 - \frac{4}{\sqrt{\pi}} \frac{R}{D}\right) \quad (5)$$

In LAC, where c is positive (typically $2-6 \text{ nm}^{-1}$), elution volumes are given by

$$V_e = V_0^* + \frac{2V_p}{cD} \exp(c^2 R^2) [1 + \text{erf}(cR)] \quad (6)$$

wherein $\text{erf}(cR)$ is the error function, and V_0^* is the accessible volume, which can be entered by a polymer chain [27, 29]:

$$V_0^* = V_i + V_p \left(1 - \frac{4R}{D\sqrt{\pi}}\right) - \frac{2V_p}{cD} = V_{\text{SEC}} - \frac{2V_p}{cD} \quad (7)$$

The radius of gyration R can be expressed by the length a and the number n of repeat units:

$$R = a\sqrt{\frac{n}{6}} \quad (8)$$

At the transition point of these two mechanisms (the CAP), the interaction parameter equals zero, and the polymer chain becomes chromatographically invisible [19, 38–40]: homopolymers without functional groups elute – independent of their molar mass – at the void volume of the column (as $K = 1$).

In the case of mono- and difunctional polymers an effective interaction parameter q accounts for the interaction of (point-type) end-groups [31]:

$$q_A = \frac{\delta}{D} [\exp(-\Delta\varepsilon) - 1] \quad (9)$$

wherein the parameter $\Delta\varepsilon$ denotes the difference in the interaction free energy (expressed in kT units) between the end group and the repeat unit; δ is the effective radius of interaction, and D is the pore diameter of the stationary phase.

Under critical conditions for the repeat unit of the polymer chain ($c = 0$) the distribution coefficient of mono-functional polymers with one end group E is given by

$$K^{(E)} = 1 + q_E \quad (10)$$

In the case of difunctionals an additional term appears: the distribution coefficient $K^{(E,E)}$ for symmetrical difunctionals (with two end groups E) is equal to

$$K^{(E,E)} \approx 1 + 2q_E + q_E^2 \frac{D}{\sqrt{\pi}R} \quad (11)$$

while asymmetrical difunctionals – with different end groups E and E' – give

$$K^{(E,E')} \approx 1 + q_E + q_{E'} + q_E q_{E'} \frac{D}{\sqrt{\pi}R} \quad (12)$$

These theoretical predictions for the chromatographic behavior of mono- and difunctional polymers have been verified in previous papers [31, 41]. The same is true for block copolymers: at the CAP for the block B of a diblock copolymer A–B or for the terminal blocks of a triblock copolymer B–A–B, the distribution coefficients of such copolymers and that of the (nonfunctional) homopolymer A are equal [19, 42]

$$K_{A-B} = K_{B-A-B} = K_A \quad (13)$$

If the interaction of the block A is strongly attractive ($c_A > 1/R_A$), the effective q_a parameter for a copolymer can be expressed as a function of both interactive and molecular parameters of the block A:

$$\begin{aligned} q_a &= K_A - 1 \approx \frac{2}{c_A D} \left[\exp(c_A^2 R_A^2) - \frac{C_A R_A}{\sqrt{\pi}} \right] \\ &\approx \frac{2 \exp(c_A^2 R_A^2)}{c_A D} \end{aligned} \quad (14)$$

In Equation 14 the definition of q_a is similar to that for the model with point-type end-groups: instead of $\Delta\varepsilon$, however, stands now the conformational free energy of an adsorbing block, which is equal to $c_A^2 R_A^2$ [43], and the former $\delta/2$ is substituted by the average thickness of the adsorbed chain A, c_A^{-1} .

At the CAP for B ($c_B = 0$) and $c_A > 0$, a B–A–B triblock can be separated exclusively according to the length of the block A, while under the same conditions the retention of an A–B–A triblock will not only depend on the length of the block B, but also on the size of the block A.

This dependence can even be utilized to determine the length of the block A, if the terminal block B is monodisperse: fatty acid diesters of PEG can very well be separated according to the number of EO units [44, 45].

Applied to the 2-D separation of EO–PO–EO and PO–EO–PO triblocks this means, that the first dimension should be a separation at the CAP for the terminal blocks. Consequently, it was necessary to find a CAP for both blocks in systems, in which the value of the interaction parameter of the other block is in a reasonable range to allow a separation of this block by the LAC mechanism.

3 Experimental

3.1 Materials

PEG, PEG–MME and PPG with different molar mass as well as some of the triblock copolymers were purchased from Fluka (Buchs, Switzerland). Di- and triblock copolymers as well as monoallyl PPG were synthesized as described previously [8, 9].

The HPLC solvents (ACN and water, both HPLC grade) were purchased from Roth (Karlsruhe, Germany). PEG and PPG standards were purchased from Sigma–Aldrich and FLUKA (Buchs, Switzerland).

3.2 Chromatography

These investigations were performed using the density detector (according to the mechanical oscillator principle [46, 47]) DDS70 (Chromtech, Graz, Austria). Data acquisition and processing were performed using the software Chroma, which has been developed for the DDS70.

In RP LC, the mobile phase was delivered by a JASCO 880 PU pump (from Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of 0.5 mL/min. Samples were injected manually using a Rheodyne 7125 injection valve

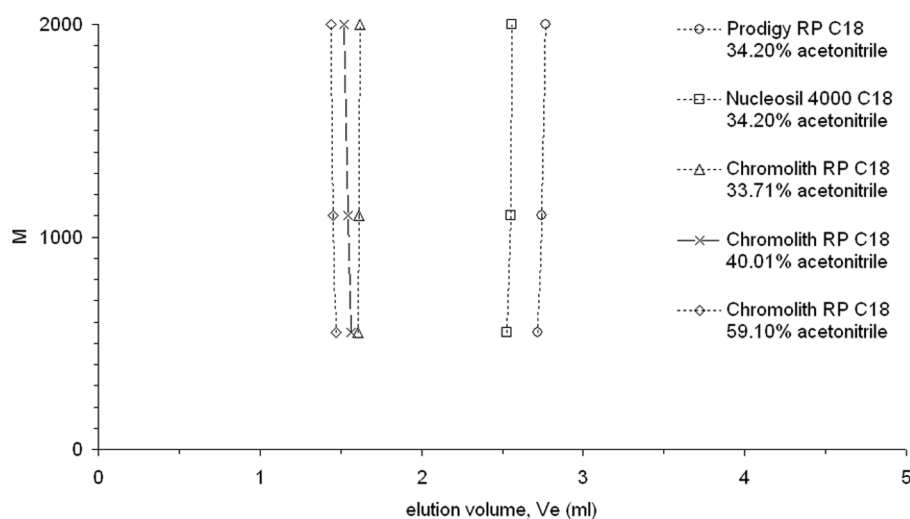


Figure 1. CAP for the EO unit on different RP columns in ACN–water mobile phases.

(from Rheodyne, Cotati, CA, USA) equipped with an 50 μ L loop. A Waters 2414 refractive index detector (Waters, Milford, MA, USA) was connected to the DDS 70. Columns were connected to two column selection valves (Rheodyne 7060, from Rheodyne).

In normal phase LC, the mobile phase was delivered by an ISCO 2350 HPLC pump (ISCO, Lincoln, NE, USA). The flow rate was 0.5 mL/min in all measurements. Samples were injected using a VICI injector equipped with an 50 μ L sample loop. A SEDEX 45 ELSD (Sedere, France) was connected to the DDS 70. Nitrogen was used as carrier gas, and the pressure at the nebulizer was set to 1.0 bar. The temperature of the evaporator was 30°C. The columns and density cells of both systems (A and B) were placed in a thermostatted box, in which a temperature of 25.0°C was maintained for all measurements using a thermostat (Lauda RM6, Lauda-Königshofen, Germany).

Mobile phases were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602 M (A. Paar, Graz, Austria).

The following columns were used in this study (specifications given by the producer):

- (i) Prodigy ODS3 (Phenomenex, Torrance, CA, USA): silica-based octadecyl phase; 250 mm $\times$ 4.6 mm; particle diameter = 5 μ m; pore size = 100 Å
- (ii) Nucleosil 4000-7 C18 (Macherey-Nagel, Germany): silica-based octadecyl phase; 250 mm $\times$ 4 mm; particle diameter = 7 μ m; pore size = 4000 Å
- (iii) Chromolith Performance RP-18e 100-4.6 (Merck, Darmstadt, Germany): silica-based monolith with octadecyl groups, 100 mm $\times$ 4.6 mm, pore size: 130 Å
- (iv) Spherisorb S5X (PhaseSeparations) plain silica; 250 $\times$ 4.6 mm; particle diameter = 5 μ m; pore size = 300 Å

- (v) Nucleosil Si 120 (Macherey-Nagel, Germany): plain silica; 250 mm $\times$ 4 mm; particle diameter = 5 μ m; pore size = 120 Å

4 Results and discussion

4.1 Reversed phases

In the literature, critical conditions of PO have been described on C18 columns and EO elutes in exclusion regime in almost all cases [24, 25, 48].

Critical conditions for EO have been observed on RP columns in methanol–water (with 80–90 wt% methanol) [5, 49] and acetone–water (with 25–30 wt% acetone) [31, 32]. In such mobile phases, PO elutes in the LAC mode [17, 30]. In acetone–water one may separate only very low oligomers of PO, as the interaction parameter of PO is too large. In methanol–water, a separation of longer PO chains can be achieved. We have now tried to find a CAP for EO in another mobile phase, which would allow a better separation according to the number of PO units.

A separation according to the number of PO unit by LAC under critical conditions for EO has also been described on RP columns in ACN–water [26]. For comparison, we have also determined the CAP on different RP columns in ACN–water.

As can be seen in Fig. 1, critical conditions were found on different columns at about 34 wt% ACN. It is also shown in the same figure that CAP for EO on such RP columns is not a sharp point but more or less a critical adsorption range (34–59% ACN). A similar behavior has already been observed on RP columns in other mobile phases: while in acetone–water there is a sharp critical point for PEG, in methanol–water a similar plateau is

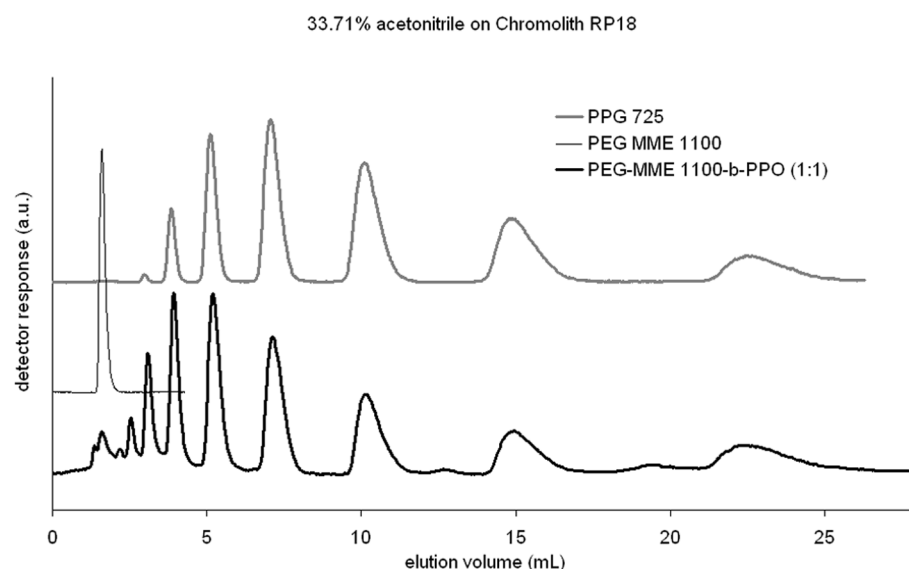


Figure 2. Overlay of chromatograms of PPG 725, PEG–MME 1100, and an EO–PO diblock copolymer based on PEG–MME 1100 at CAP of EO, as obtained on the Chromolith RP18 column, ELSD.

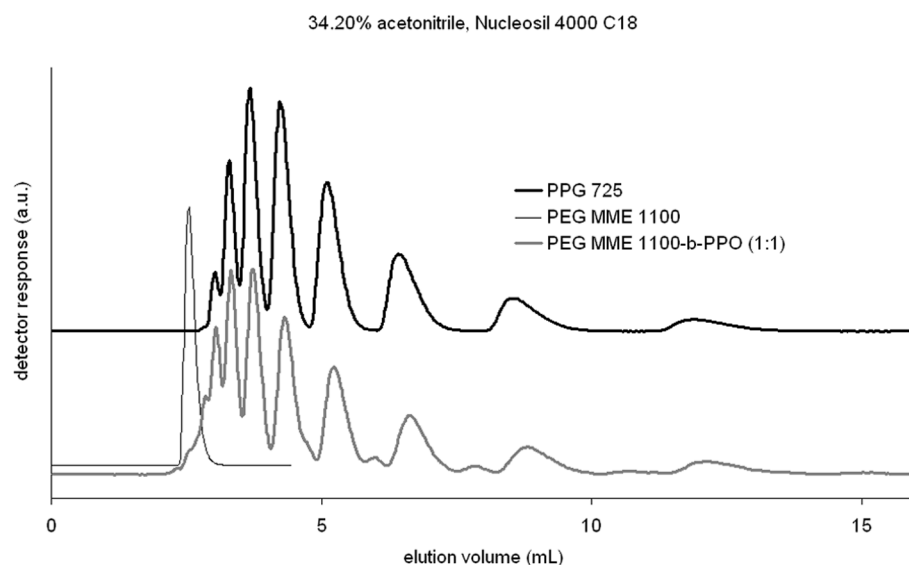


Figure 3. Overlay of chromatograms of PPG 725, PEG–MME 1100, and an EO–PO diblock copolymer based on PEG–MME 1100 at CAP of EO: Detection, ELSD.

observed [32], even though it is not as wide as in ACN–water.

In such a mobile phase, EO–PO diblocks elute just like the corresponding PPGs: on the Chromolith RP18 column (Fig. 2) retention (and resolution) is higher than on the macroporous Nucleosil 4000 C18 column (Fig. 3). As the critical point is rather a critical range in ACN–water (similar to methanol–water), one may adjust the retention not only by choosing a column with different pore dimensions, but also by moderate variation of mobile phase composition.

The dependence of the interaction parameter of PO on the Chromolith column in this range of mobile phase composition is shown in Fig. 4.

A similar separation can be achieved with EO–PO–EO triblocks, as can be seen in Fig. 5.

While the separation according to the number of PO units at the CAP for EO can be considered to be solved, the main goal of this study was to find a system on which PO is critical while EO elutes in LAC mode.

4.2 Normal phases

The first step was to find a CAP for PO on normal phase columns. This could be realized on plain silica columns in ACN–water mobile phases. We have used silica columns from two different producers (Nucleosil from Macherey-Nagel and Spherisorb from PhaseSeparations), on which critical conditions can be realized at different mobile phase compositions. These columns should have the same chemistry and thus the same activity, which is, however, not the case. The different behavior could be

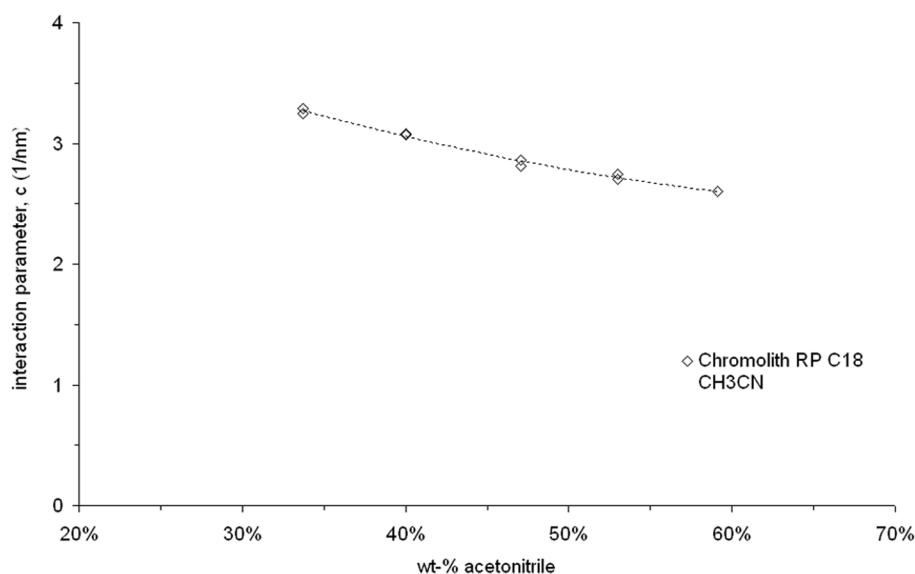


Figure 4. Interaction parameter c of PPG on the Chromolith column in ACN–water mobile phases of different composition.

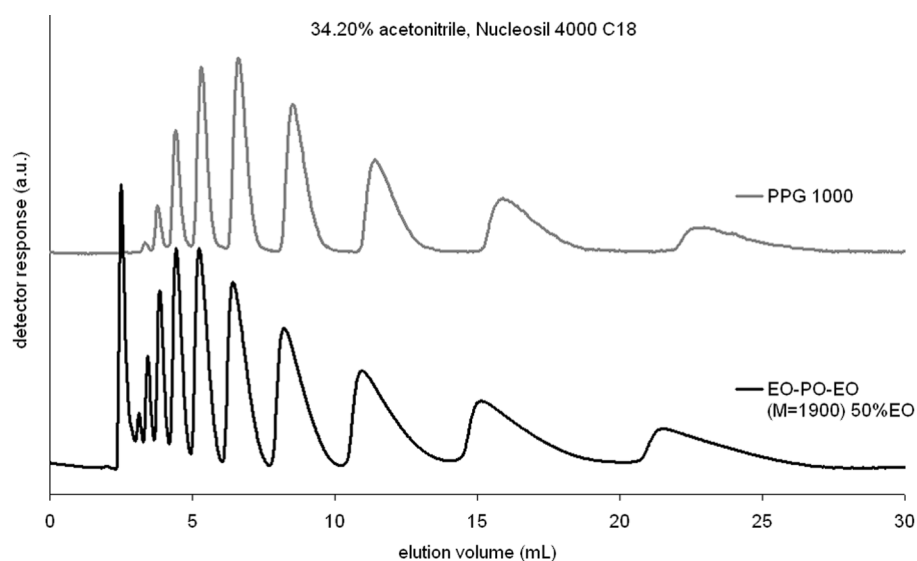


Figure 5. Overlay of chromatograms of PPG 1000 and a commercial EO–PO–EO block copolymer with a molar mass of 1900 and 50% EO at CAP of EO. Detection: RI for block copolymer and ELSD for homopolymer.

due to a different number of silanol groups and ether linkages on the surface of the stationary phase.

On the Spherisorb S5X column a CAP for PO was found at about 71 wt% ACN (Fig. 6). Under these conditions PEG has a positive interaction parameter (approximately 2 nm^{-1}) and thus elutes in LAC order. On this column, which has a pore diameter of $300 \text{ \AA}$, we could separate the homopolymer (PPG) from the block copolymer and furthermore we can separate the block copolymer oligomers according to the length of the EO block irrespective of the PO block. As can be seen in Fig. 7, the peak positions of the individual oligomers of EO in the block copolymers and the initiator (PEG MME 1100) are the same.

A good resolution can also be achieved in a slightly different mobile phase composition, which is not exactly critical (Fig. 8). The difference in mobile phase composi-

tion is small (about 1%), but the resolution is considerably improved.

On the Nucleosil Si $120 \text{ \AA}$ column, we found a CAP for PO at around 87 wt% ACN (see Fig. 9), which is quite different from that on the Spherisorb S5X column. In such a mobile phase, PEG has an interaction parameter of approximately 2.4 nm^{-1} .

As is shown in Fig. 10, PPG diols could be separated from PPO with allylic end group on the Nucleosil $120 \text{ \AA}$ Si column. PPG diols elute at 3.4 mL while PPO with an allylic end group elute sufficiently earlier (2.8 mL) to allow a baseline separation. It is evident that commercial PPG 3000 contains some traces of polypropylene oxide with allylic end group.

Under these conditions PEG elutes in LAC mode. By using this mobile phase composition (87 wt% ACN) on narrow pore column ($120 \text{ \AA}$), a block copolymer could be

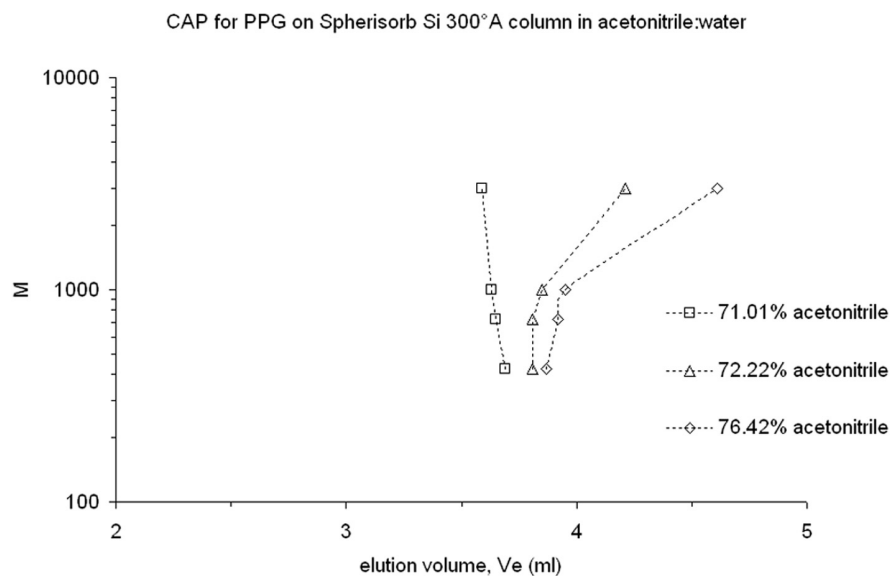


Figure 6. Critical conditions for PPG on the Spherisorb S5X column (300 Å).

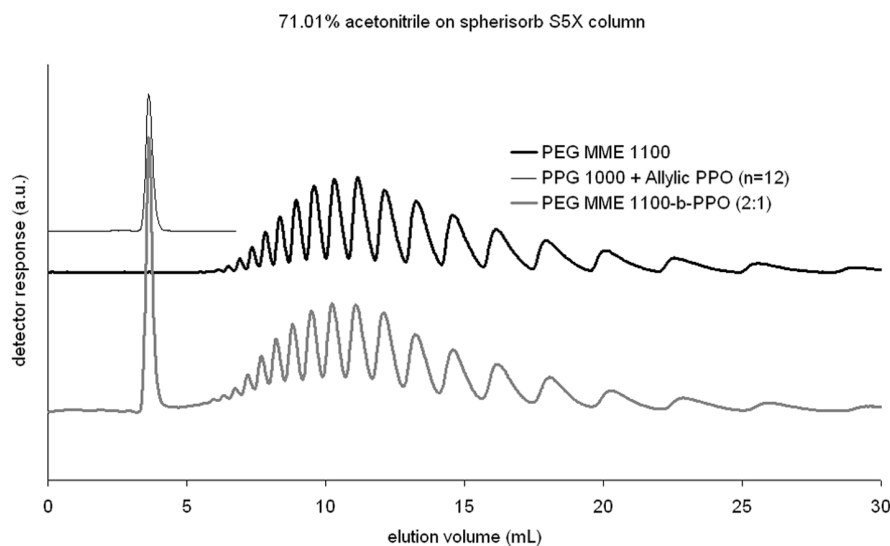


Figure 7. Overlay of chromatograms of PEG MME 1100, EO–PO block copolymer and PPG (PPG 1000 + PPO with allylic end group) at the CAP of PO (ELSD).

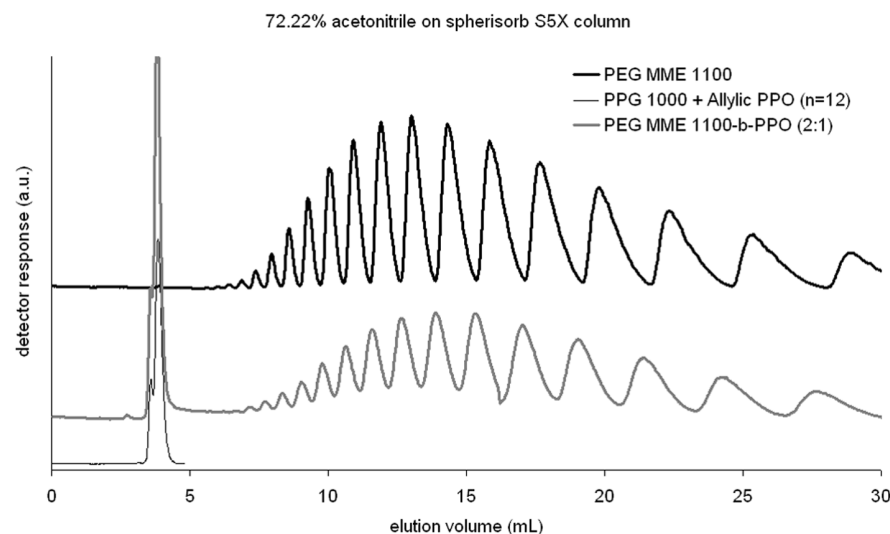


Figure 8. Overlay of chromatograms of PEG MME 1100, EO–PO block copolymer and PPG (PPG 1000 + PPO with allylic end group) at the CAP of PO (ELSD).

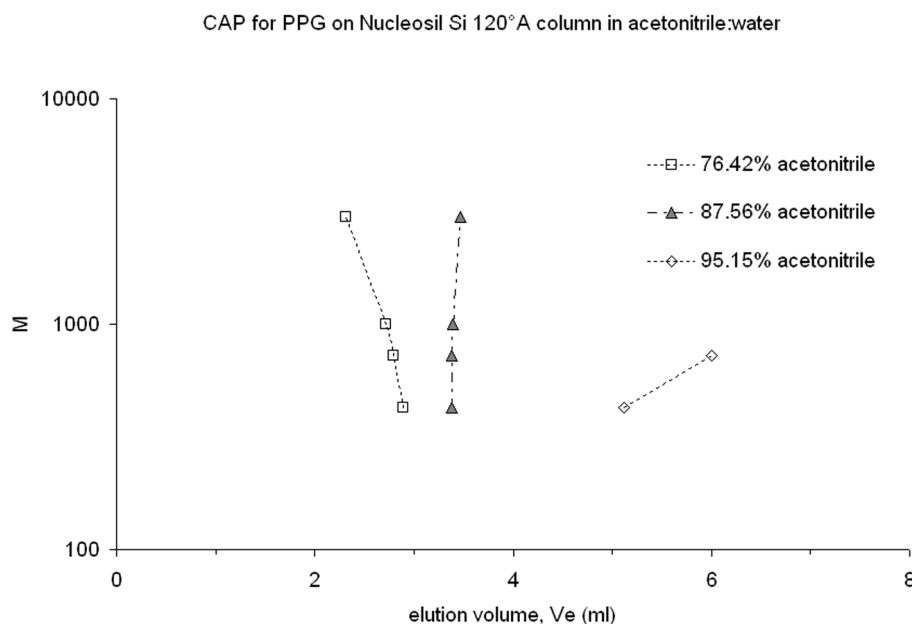


Figure 9. Critical conditions for PPG on Nucleosil Si 120 Å column.

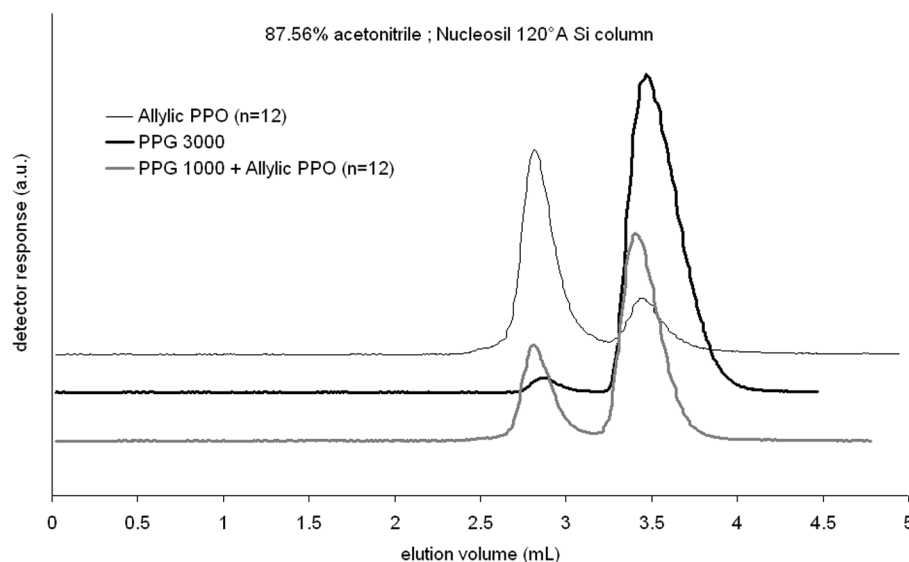


Figure 10. Separation of PPG diols from PPO with allylic end group at the CAP for PO on the Nucleosil Si 120 Å column; ELSD.

separated with respect to the length of the PEG block. If there is any PO homopolymer formed, it should elute earlier at void volume of the column.

On this column the individual oligomers of PEG–MME 550 can be separated to the baseline by LAC, and the same is true for the EO block of an EO–PO diblock copolymer, which is based on PEG–MME 550 as an initiator (Fig. 11). Peak positions of initiator (PEG MME 550) and block copolymer are exactly same. Obviously, this sample also contains traces of PPG diol and PPO homopolymer with allylic end group.

An even better separation can be achieved with a PO–EO–PO triblock copolymer based on PEG 300 as initiator

(Fig. 12). In the chromatogram of the initiator the density trace is shown, as the ELSD does not detect the lowest oligomers of PEG [50] (see Fig. 13).

Figure 14 shows a similar separation of a PO–EO–PO triblock copolymer based on PEG 600 as an initiator. This sample also contains some PPG diol and PPG monoallyl ether. The resolution of the individual oligomers is even better in this case, and the peak positions of the PEG homopolymer and the block copolymer are almost the same. The applications of above findings in offline and comprehensive (online) 2-D chromatography of such block copolymers is already under investigation and shall be described in a forthcoming publication.

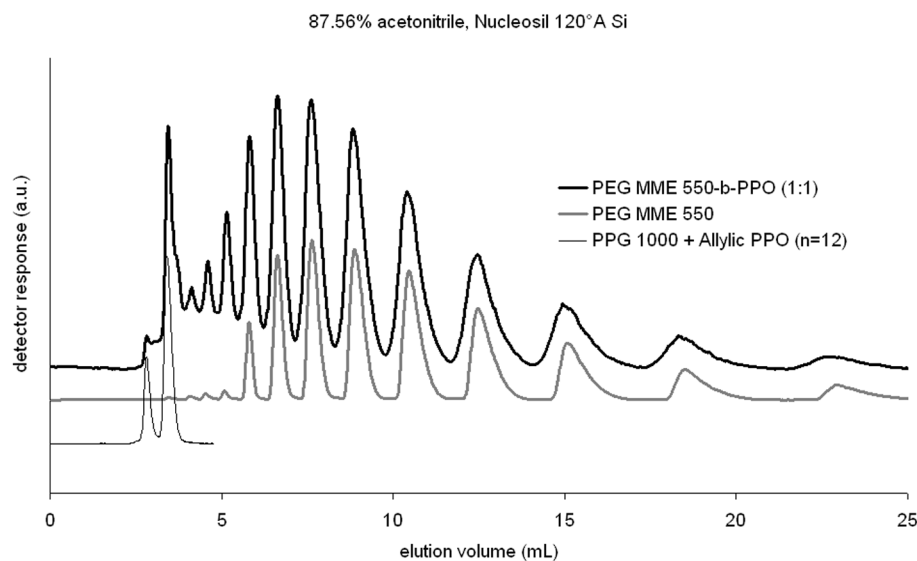


Figure 11. Overlay of chromatograms of PPG (PPG 1000 + PPO with allylic end group), PEG–MME 550 and block copolymer of PEG MME 550-*b*-PPO (1:1) at the CAP for PO on Nucleosil Si 120 Å column, ELSD.

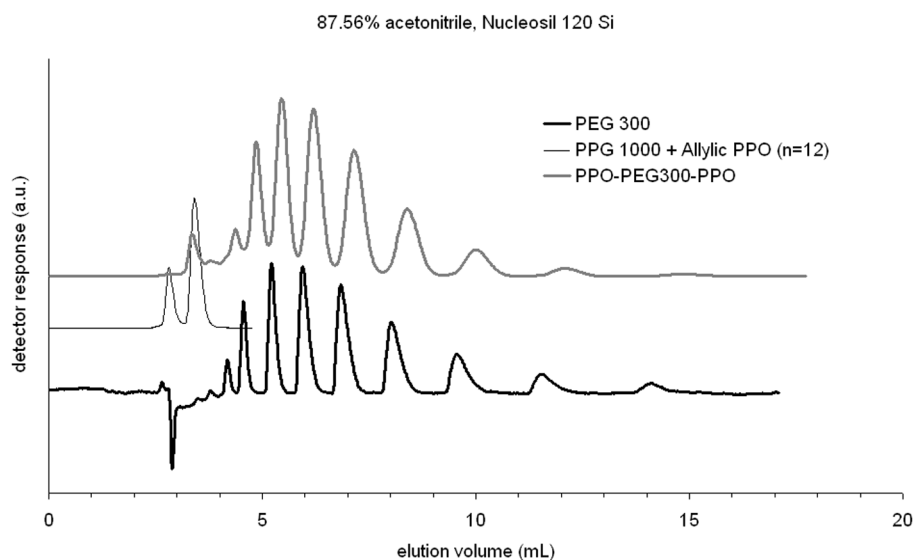


Figure 12. Overlay of PPG (PPG 1000 + allyl-PPO), PEG 300 and a triblock copolymer PPO-*b*-PEG300-*b*-PPO (1:1) at the CAP for PO on the Nucleosil Si 120 Å column. Detection: ELSD for PPG and block copolymer, density for PEG 300.

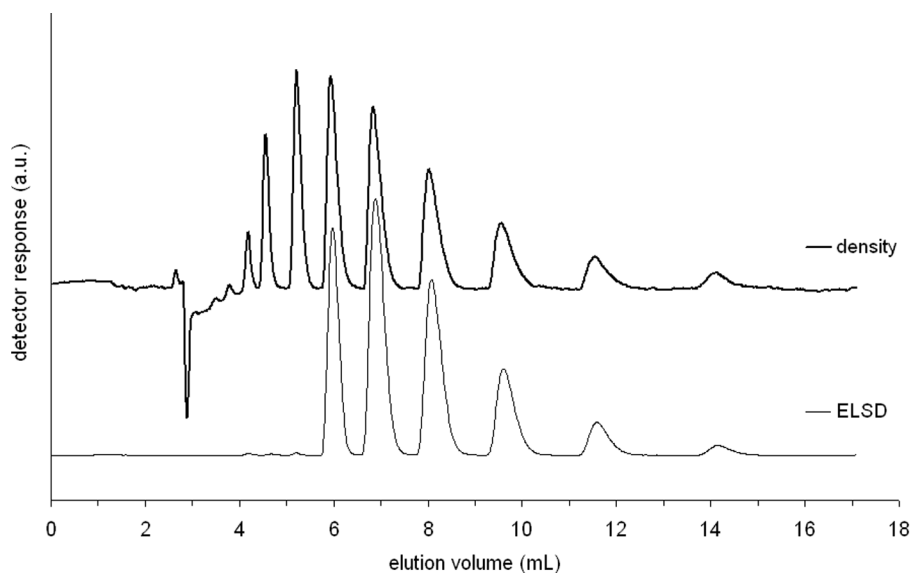


Figure 13. LAC of PEG 300 on the Nucleosil Si 120 Å column in 87.56 wt% ACN. Detection, density and ELSD.

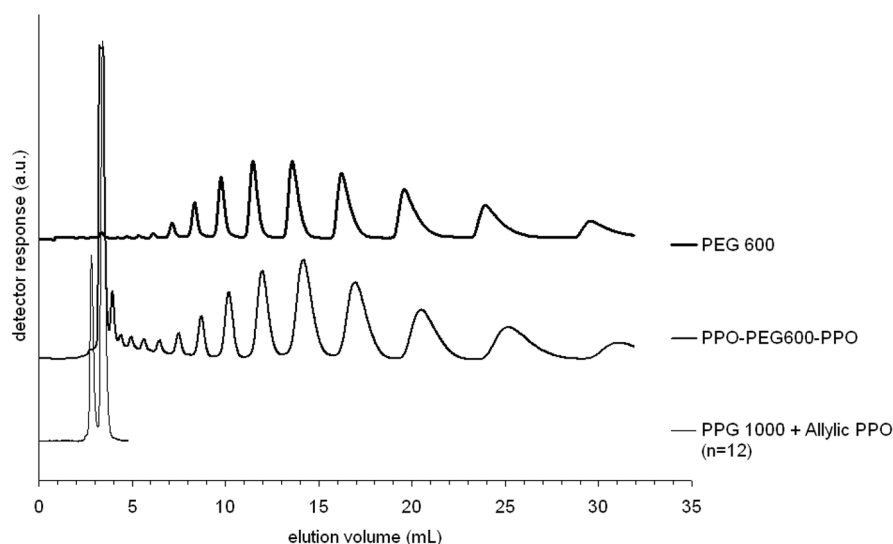


Figure 14. Overlay of PPG (PPG 1000 + allyl-PPO), PEG 600 and a triblock copolymer PPO-*b*-PEG600-*b*-PPO (1:1) at the CAP for PO on the Nucleosil Si 120 Å column, ELSD.

5 Conclusions

EO-PO diblock copolymers are analyzed by using LCCC. In this report, we used CAP for one block and adsorption for the second block. Depending on the pore diameter of the column used, all oligomers with respect to both blocks can be resolved within reasonable range of molecular weight (around 1000–1500 for each block).

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